

Muhammad Mubashir

+92-333-7588703 | mmubashirmemon@gmail.com | [linkedin.com/in/mmubashir72](https://www.linkedin.com/in/mmubashir72) | github.com/mubashir72

SUMMARY

MS Artificial Intelligence student at FAST NUCES with hands-on experience in AI Agents, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Python-based AI applications. Experienced in extending AI research systems, developing RAG applications, and rapidly prototyping AI solutions during hackathons. Passionate about building practical AI systems using LangChain, OpenAI APIs, and modern LLM frameworks.

EDUCATION

FAST NUCES

Master of Science in Artificial Intelligence, CGPA: 3.44/4.00

Islamabad, PK

Aug. 2025 – Present

MUET

Bachelor's of Engineering in Software Engineering

Jamshoro, PK

Nov. 2020 – Nov 2024

EXPERIENCE

Backend AI Engineering Intern

Aug 2026 – Present

FlyRank.AI

Remote

- Developed backend APIs and asynchronous microservices using Python and FastAPI.
- Engineered clean architecture using the repository pattern to decouple data access layers from core business logic and API routes.
- Currently configuring multi-container environments using Docker and Docker Compose to integrate persistent PostgreSQL databases with volume management.

AI Engineering Intern

July 2026

Brinicle AI

Remote

- Worked on AI engineering tasks involving LLMs, generative AI, and practical AI application development.
- Conducted technical experiments, documented findings, and strengthened practical AI/ML development skills.

WordPress Developer Intern

July 2025 – Aug. 2025

KGT Global

Karachi

- Developed and customized responsive WordPress websites using Divi Builder, Elementor, PHP, JavaScript, HTML, and CSS.
- Implemented features, resolved bugs, and optimized website performance and user experience.

PROJECTS

YouTube Video Q&A System using RAG | *Python, OpenAI, Pinecone, Whisper, Langchain, Gradio*

- Developed an end-to-end RAG application enabling users to query YouTube videos through natural language using video URLs.
- Built transcription extraction, chunking, embedding, and semantic retrieval pipelines with OpenAI Embeddings and Pinecone.
- Integrated LangChain-based retriever + LLM workflow for context-aware question answering.
- Designed an interactive Gradio interface with video input, thumbnail preview, and real-time response generation.
- Implemented caching and vector store optimization to improve retrieval speed and reduce redundant processing.

Brain Tumor Classification using CNN | *Python, Tensorflow*

- Developed a CNN-based deep learning model using TensorFlow and Keras to classify brain MRI images into glioma, meningioma, and pituitary tumor classes.
- Achieved 95% accuracy on a dataset of 19,000+ labeled images using data augmentation and architecture tuning.
- Performed data preprocessing, visualization, and evaluation using Python, NumPy, Matplotlib, scikit-learn, and TensorFlow.
- Deployed trained model for inference and performance evaluation on unseen test data.

Medical Chest X-ray Analysis Agent | *Python, PyTorch, LLMs, Computer Vision*

- Extended an existing medical AI agent framework for chest X-ray analysis as part of an ongoing research project.
- Integrated Grad-CAM explainability into the diagnosis pipeline to improve model interpretability.
- Evaluated and compared multiple Large Language Models to improve diagnostic report generation quality.
- Modified and extended the existing codebase to support research experiments and benchmarking.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, Java, HTML, CSS, JavaScript, Tailwind CSS, React, MongoDB, FastAPI
Machine Learning & AI: TensorFlow, PyTorch, Scikit-Learn, LangChain, OpenAI API, Retrieval-Augmented Generation (RAG), Prompt Engineering, Pandas, NumPy, Matplotlib
Frameworks & Tools: Gradio, Git, VS Code, PyCharm, Visual Studio, Eclipse
Soft Skills: Problem-Solving, Communication Skills, Quick Learner, Team Collaboration, Time Management, Adaptability, Attention to Detail, Adaptability

CERTIFICATIONS

Machine Learning Specialization DeepLearning.AI — Instructor: Andrew Ng (Includes: Supervised Machine Learning, Advanced Learning Algorithms, and Unsupervised Learning, Recommenders & Reinforcement Learning)	Coursera
PyTorch: Fundamentals DeepLearning.AI — Instructor: Laurence Moroney	Coursera
Introduction to Web Development with HTML, CSS, JavaScript IBM — Instructor: Upkar Lidder, Rav Ahuja, Michelle Saltoun	Coursera
Fundamentals of Building AI Agents IBM — Instructor: JOSEPH SANTARCANGELO, Faranak Heidari, Wojciech Fulmyk, Karan Goswami	Coursera